

RMO(INDIA) – 2006

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be an acute angled triangle and let D, E, F be the foot of perpendiculars from A, B, C respectively to BC, CA, AB. Let the perpendiculars from F to CB, CA, AD, BE meets them in P, Q, M, N respectively. Prove that P, Q, M, N are collinear.
 2. Find the least possible value of $a + b$, where a, b are positive integers such that 11 divides $a + 13b$ and 13 divides $a + 11b$.
 3. If a, b, c are three positive real numbers, prove that: $\frac{a^2 + 1}{b + c} + \frac{b^2 + 1}{c + a} + \frac{c^2 + 1}{a + b} \geq 3$.
 4. A 6X6 square is dissected into 9 rectangles by lines parallel to its sides such that all these rectangles have only integer sides. Prove that there are always two congruent rectangles.
 5. Let ABCD be a quadrilateral in which AB is parallel to CD and perpendicular to AD; $AB = 3CD$ and the area of the quadrilateral is 4. If a circle can be drawn touching all the sides of the quadrilateral. Find its radius.
 6. Prove that there are infinitely many positive integers n, such that $n(n+1)$ can be expressed as a sum of two positive squares in at least two different ways. (Here $a^2 + b^2$ and $b^2 + a^2$ are considered as the same representations).
 7. Let X be the set of all positive integers greater than or equal to 8 and let $f : X \rightarrow X$ be a function such that $f(x + y) = f(xy)$ for all $x \geq 4, y \geq 4$. If $f(8) = 9$, determine $f(9)$.

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